

Strongly Non-Repetitive Sequences and Progression-Free Sets

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By a simple method we show the existence of (1) a sequence on two symbols in which no four blocks occur consecutively that are permutations of each other, and (2) a sequence on three symbols in which no three blocks occur consecutively that are permutations of each other. The problem of the existence of a sequence on four symbols in which no two blocks occur consecutively that are permutations of each other remains open.

1. INTRODUCTION

Is there a sequence on four symbols in which no two adjacent segments are permutations of one another? We do not know the answer to this question. In this paper we present a method which might lead to a solution and which solves two closely related problems posed in [1].

Let I be a finite set. A *block* B over I of *length* m is a finite sequence $B = b_1 \cdots b_m$ with $b_k \in I$. We say that B *occurs* at place p in an infinite sequence $x = x_1 x_2 \cdots$ over I , if $x_p x_{p+1} \cdots x_{p+m-1} = B$; blocks $B_1, B_2, \dots, B_n$ occur *consecutively* in x , if the block $B_1 B_2 \cdots B_n$ (juxtaposition of $B_1, B_2, \dots, B_n$) occurs in x .

A *strongly non-repetitive* sequence x of *order* n is an infinite sequence such that no n blocks $B_1, \dots, B_n$ occur consecutively in x which are permutations of each other.

The problem of the existence of a strongly non-repetitive sequence of order n over a set I of cardinality r will be called the (r, n) -problem. The $(25, 2)$ -problem was solved in 1968 by Evdokimov [3]. Pleasants [5] drastically lowered the number of symbols by solving the $(5, 2)$ -problem two years later. On the other hand the $(2, 5)$ -problem was solved by Justin [4].

It is easily checked that no solutions to the $(3, 2)$ - and $(2, 3)$ -problems exist. Considering the afore mentioned results this means that the only remaining non-trivial cases are the $(4, 2)$ -, $(3, 3)$ - and $(2, 4)$ -problems. We shall give simple solutions to the latter two by a method which generalizes ideas emerging from [3] and [4], and which is, rather unexpectedly, not altogether strange to the solution in [5].

2. PRELIMINARIES

The *frequency vector* of a block B is the row vector with as i th component the number of times that $i \in I$ occurs in B . Note that two blocks are permutations of each other iff their frequency vectors are equal.

In solving problems of the type encountered in the introduction, substitution is an ubiquitous tool [3, 4, 5]. A *substitution* θ is a map from I to the set of all blocks over I . The blocks θi with $i \in I$ will be called θ -blocks. We shall only consider substitutions such that all θ -blocks have length at least 2. The matrix of a substitution θ is the matrix whose j th row is the frequency vector of θj .

Let E denote the empty block. If $i \in I$ and $\theta i = VV'$ is any decomposition of θi with $V' \neq E$ then we call V a *left subblock* of θ .

The domain of a substitution extends to blocks by setting

$$\theta(b_1 b_2 \cdots b_m) = \theta b_1 \theta b_2 \cdots \theta b_m.$$

If for some $j \in I$ the block θj starts with j then we may define an infinite sequence x by repeatedly applying θ to j . We call x a sequence *generated* by θ . (In this formulation we suppressed j , since the properties of x in which we are interested do not depend on a particular choice of j).

In our set up we also need a finite Abelian group G (with neutral element 0) and a map $f: I \rightarrow G$ whose domain extends to the set of all blocks by defining

$$f(b_1 b_2 \cdots b_m) = f(b_1) + f(b_2) + \cdots + f(b_m), \quad f(E) = 0.$$

Given θ, f and G we call f θ -*injective*, if for all n and any collection $V_1, \dots, V_n$ of left subblocks of θ the equality

$$f(V_1) = \cdots = f(V_n)$$

implies either $V_1 = \cdots = V_n$ or $V'_1 = \cdots = V'_n$.

Finally we call a subset A of an Abelian group G *progression-free* of order n if

$$a \in A, \quad a + g \in A, \dots, \quad a + (n-1)g \in A$$

implies $g = 0$. (This notion has been introduced for subsets of $\mathbf{Z}$ in [2]. Here we also find the following basic example. Take $m \geq 1$, $G = \bigoplus^m \mathbf{Z}/3\mathbf{Z}$, then $A = \{(g_1, \dots, g_m) \in G: g_k \in \{0, 1\}\}$ is a progression-free set of order 3).

LEMMA. Let $n > 1$, θ a substitution on a set I , G a finite Abelian group and f a map $f: I \rightarrow G$ such that

- (i) the matrix of θ is non-singular,
- (ii) for each $i \in I$, $f(\theta i) = 0$,

(iii) the set $A = \{g \in G: g = f(V), V \text{ any left subblock of } \theta\}$ is progression free of order $n + 1$,

(iv) f is θ -injective.

Then any sequence generated by θ is strongly non-repetitive of order n .

Proof. Let x be a sequence generated by θ . Let the length of B_1 be minimal such that $B_1, B_2, \dots, B_n$ occur consecutively in x , with the B_k permutations of each other. Each B_k starts in some θ -block θi_k and B_n ends in some θ -block θi_{n+1} . Let V_k be the corresponding left subblock, i.e., $\theta i_k = V_k V'_k$, and B_k occurs at the same place as V'_k in x for $1 \leq k \leq n$, B_n ends at the same place as V_{n+1} in x . We take $V_k = E$ if B_k occurs at the same place as θi_k .

It follows from $f(B_1) = f(B_2) = \dots = f(B_n)$ and (ii) that $f(V_1), f(V_2), \dots, f(V_{n+1})$ is an arithmetic sequence. Now (iii) implies

$$f(V_1) = f(V_2) = \dots = f(V_{n+1}).$$

By (iv) therefore, we can find a block $C_1 C_2 \dots C_n$ in x , with the C_k permutations of each other and each C_k composed of θ -blocks. Let D_k be defined by $C_k = \theta D_k$. Then $D_1 D_2 \dots D_n$ occurs in x . Since the frequency vector of C_k can be obtained by multiplying the frequency vector of D_k with the matrix of θ , (i) implies that the D_k are permutations of each other. This contradicts the minimality of the length of B_1 .

3. RESULTS

THEOREM 1. *There exists a sequence on two symbols in which no four blocks occur consecutively that are permutations of each other.*

Proof. Let $I = \{a, b\}$, and θ be defined by

$$\theta a = abb,$$

$$\theta b = aaab.$$

The matrix $\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ of θ is non-singular. Choose $G = \mathbb{Z}/5\mathbb{Z}$ and define $f: I \rightarrow G$ by

$$f(a) = 1, \quad f(b) = 2.$$

Then

$$(f(a), f(ab), f(abb)) = (1, 3, 0),$$

$$(f(a), f(aa), f(aaa), f(aaab)) = (1, 2, 3, 0).$$

From this we immediately see that $f(\theta a) = f(\theta b) = 0$, that f is θ -injective and that

$$A := \{g \in G: g = f(V), V \text{ any left subblock of } \theta\} = \{0, 1, 2, 3\}.$$

Since A is a progression-free set of order 5 in G , the theorem follows by the lemma of Section 2, taking for x the sequence obtained by repeatedly applying θ to a .

THEOREM 2. *There exists a sequence on three symbols in which no three blocks occur consecutively that are permutations of each other.*

Proof. Let $I = \{a, b, c\}$, and θ be defined by

$$\theta a = aabc,$$

$$\theta b = bbc,$$

$$\theta c = acc.$$

Choose $G = \mathbf{Z}/7\mathbf{Z}$ and define $f: I \rightarrow G$ by

$$f(a) = 1, \quad f(b) = 2, \quad f(c) = 3.$$

As in the proof of Theorem 1 it is a simple task to check the conditions of the lemma of Section 2 with $n = 3$.

4. ANOTHER FOUR-COLOR PROBLEM?

Although the lemma of Section 2 yields simple solutions to the (2, 4)- and (3, 3)-problems, we have not been able to solve the (4, 2)-problem. The following observation suggests that the latter might be much more difficult to tackle. Pleasants' result, the solution of the (5, 2)-problem, allows a proof in the spirit of our lemma, with the same substitution as his, although his methods are quite different. Specifically, there exists a group G and a map $f: \{a, b, c, d, e\} \rightarrow G$ nearly satisfying property (iii), and fully satisfying properties (ii) and (iv). However the group G in this case has order 9375.

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